22.2. RADIONUCLIDES OF ENVIRONMENTAL CONCERN

Most of the radionuclides produced in nuclear tests, accidents and in the normal fuel cycle are short lived. In Table 21.3, longer lived fission products and activation products from these systems are listed as these represent the major concern to the general public if they are allowed to enter the environment; we exclude nuclides with insignificant contribution to the total activity after 10 years. In addition, several heavy radionuclides are formed by neutron capture reactions such as ^{236}U, ^{237}Np, $^{238-242}$Pu, ^{241}Am, etc.

To assess the potential for these radionuclides to cause harm to humans, their geochemical and biological behavior must be evaluated. For example, since Kr is a chemically inert gaseous element, it would have little effect on a person who inhaled and immediately exhaled the small amount which might be present in the air. By contrast, other nuclides with high activity, ^{90}Sr–^{90}Y and ^{137}Cs–^{137m}Ba, have active geological and biological behavior and can present much more significant radiation concerns to humans. In normal operations, these nuclides would of course be released in quite insignificant amounts, as represented by the value for aquatic releases in reactor operation and reprocessing, Table 19.7 and 21.10. Similarly, the heavy elements (Np, Pu, Am) would not be released in normal operation at levels that would be of concern. We have pointed out that some low level releases as a result of normal operations are allowed by the health authorities, who also monitor these levels. In the effluents from reprocessing plants (e.g. Sellafield in the United Kingdom and La Hague in France), the relatively long-lived nuclides such as ^{3}H, ^{14}C, ^{85}Kr, ^{99}Tc and ^{129}I are of major concern. The liquid effluents from nuclear power plants and from reprocessing plants are about equally responsible for the global collective dose commitment of nuclear power generation (i.e., $\sim$0.8 man Sv per GW_ey of the total $\sim$2.5 man Sv).

22.3. RELEASES FROM LARGE REACTOR ACCIDENTS

So far there have only been two major nuclear reactor accidents, Chernobyl and Fukushima. In both cases large amounts of radioactive material was released into the environment. However, in these events there was only an insignificant addition of ^{14}C to the atmosphere relative to the average atmospheric amount of ^{14}C at the corresponding times.

22.3.1. Chernobyl

On April 26, 1986, a low power engineering test was being conducted at one of the reactors of the Chernobyl nuclear power station in the Ukraine (then the USSR). The reactor became unstable, resulting in thermal explosions and fires that caused

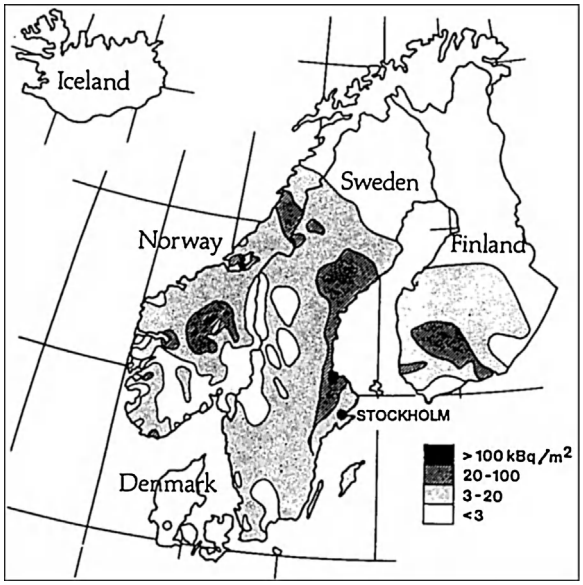


Figure 22.3 Deposition of ^{137}Cs in Scandinavia from the Chernobyl accident.

severe damage to the reactor and its building (§§13.10.2 and 20.1.2.c). Radioactivity was released over the next ten days until the fires were extinguished and the reactor entombed in concrete. The radioactivity was released as gas and dust particles and initially blown by winds in a northerly direction. Outside Russia, the accident was first detected by increased radioactivity levels at the Forsmark nuclear power plant, about 110 km north of Stockholm, Sweden, where it caused a full alarm as the radioactivity was believed to come from the Swedish plant. Subsequently, the radioactivity released at Chernobyl was spread more to the west and southwest (Figures 13.10, 22.3, and 22.4).

For the exposed population in the Byelorussia region near Chernobyl the estimated average increased dose in the first year after the accident was approximately the same as the annual background radiation. In northern and eastern Europe in general, the increased exposure during the first year was 25–75% above background levels. The highest dose will be delivered in southeastern Europe and is estimated to be 1.2 mSv up to year 2020, which can be compared to ~70 mSv from natural background radiation during the same period. Figure 22.3 shows the levels of ^{137}Cs deposited in Scandinavia in the first days after the accident. Table 22.4 gives the fraction of the core activity released and the air and ground contamination of various nuclides at two Swedish locations. Figure 22.4 shows the fallout in part of the northern hemisphere.



Figure 22.4 Deposition from Chernobyl in much of the northern hemisphere.

It was shown that the larger airborne particulates from the Chernobyl accident had a composition which was quite similar to that of the reactor fuel. A comparison of the composition of these "hot particles" with that of the reactor fuel is given in Table 22.5. About a tenth of these hot particles had a high concentration of ^{103}Ru and ^{106}Ru while others were depleted in the ruthenium fission products. The Ru-rich particles may have originated from a part of the reactor where burning graphite produced CO which reduced the ruthenium to non-volatile metallic Ru. In other sections, oxidation occurred, forming volatile RuO_3 and/or RuO_4 which vaporized from the particles.

In contrast to the larger particles, the composition of the smaller ones varied considerably and they were distributed over much greater distances. Other measurements reflect this variability in the fall-out from Chernobyl. For example, 70% of the ^{137}Cs from Chernobyl measured in Great Britain was water soluble. By contrast, the ^{137}Cs measured in Prague, much closer to the accident site, was only 30% water

Table 22.5 Nuclide composition (%) of a "hot particle" from Chernobyl compared to reactor fuel after 3 years of burning

Nuclide	"Hot Particle"	Reactor fuel
^{95}Zr	17.9	17.6
^{95}Nb	20.7	19.3
^{103}Ru	14.2	15.0
^{106}Ru(Rh)	3.5	2.4
^{140}Ba(La)	12.6	11.1
^{141}Ce	15.8	14.3
^{144}Ce(Pr)	15.8	14.4

soluble. Further insight into the variety of species present in the Chernobyl dust is found in data on the deposition of some radionuclides during periods of rain, and dry weather. Rainy periods accounted for 70–80% of the total deposition of ^{134}Cs, ^{137}Cs, ^{103}Ru, ^{106}Ru, and ^{132}Te while deposition during dry weather was more important for ^{131}I.

These observations indicate that the speciation of radionuclides in the atmosphere is dependent on their source, their mechanisms of production and the nature of the particular environment. While some species are gaseous, others are associated to particles with properties and suspension times that are strongly dependent on the particle size and density.

22.3.2. Fukushima

On the 11th of March 2011 an earthquake of the magnitude 9.0 on the Richter scale occurred outside the eastern coast of Japan. All affected nuclear reactors shut down as planned as a response to the earthquake. However, an unusually large tsunami hit the Fukushima Daiichi nuclear power station afterwards causing a total blackout. As the fuel heated up the zircaloy cladding underwent a highly exothermic oxidation reaction according to eqn. (19.44).

$$Zr + 2H_2O \rightarrow ZrO_2 + 2H_2 \quad \Delta H = -6.67 + 2.57 \times 10^{-4}\,T\,\left(MJ/kg\,Zr\right) \quad (19.44)$$

The hydrogen produced escaped up into the secondary reactor building where it accumulated and later exploded. Such explosions occurred in two reactors, but at different times. This nuclear accident and its release was more fully described earlier, see §13.10.2.

In comparison, the total releases of ~900 PBq total into the air from Fukushima are modest when compared to the ~5200 PBq total from Chernobyl. The fallout in Japan resulted in evacuation of many of the most afflicted areas. The effect of rain on the fallout was rather similar in the two cases.

22.4. INJECTION OF TRU INTO THE ENVIRONMENT

Of the artificial radionuclides released to the environment by nuclear activities, the transuranium (TRU) species are a major concern. This concern arises from the very long half-life of a number of the nuclides as well as their high radiotoxicity values. Although reactor operation and spent fuel reprocessing activities have released small amounts of TRU's to the environment, testing of nuclear weapons has released rather large quantities. Since the first nuclear test detonation in New Mexico in 1945, approximately 3 500 kg of plutonium has been released in atmospheric explosions and another 100 kg in underground tests. This corresponds to about 11 PBq of $^{239+240}$Pu ejected into the atmosphere. In addition 0.6 PBq of ^{238}Pu were released over the south Pacific in the high altitude destruction of the SNAP-9 satellite power source in 1964. By contrast, a total of ~0.58 PBq of $^{239+240}$Pu has been released into the Irish Sea from the Sellafield (UK) reprocessing plants between 1971 and 1999; most of this before 1985. About 37 kg of ^{241}Am is present in the environment from the decay of ^{241}Pu from the nuclear testing. Table 22.6 compares the TRU's released in nuclear tests in the atmosphere, in the Irish Sea from the Sellafield plants, and into the English Channel from the LaHague plants. As the amount of spent nuclear fuel increases, the contribution to the total plutonium in the environment could become more significant over a longer time, especially if nuclear waste disposal sites release actinide elements slowly to the environment. Whatever the sources of plutonium and other actinides, their presence represents a contamination of the environment by highly toxic material. An understanding of the factors involved in their retention and/or migration in the ecosphere is therefore highly desirable. Studies of the environmental behavior of releases from tests provide data needed to understand and predict the behavior of smaller releases from the nuclear power industry. Therefore, in the next few paragraphs we concentrate our discussion to the behavior of the actinide elements in the environment, particulary the mobility of plutonium.

Table 22.6 TRU released from nuclear tests and from reprocessing (UNSCEAR2000, BNFL and COGEMA)

	Tests (TBq)	Sellafield (TBq)		LaHague (TBq)	
Nuclide	**1945–1980**	**1971–84**	**1985–94**	**1995–99**	**2000**
^{237}Np	–	–	0	0.33	
^{239}Pu	6520				
^{240}Pu	4350	559	15	0.92	
^{241}Pu	142000	–	–	21.8	0.039[†]
^{241}Am	–	442	9	0.31	
^{242}Cm	–	–	–	0.052	
$^{243+244}$Cm	–	–	–	0.024	

[†]Total α.